

Exercise 05

PART A

For each of the following questions there are four choices marked A, B, C and D. Only one of them is correct. You should decide on the correct choice and write down its mark on the answer sheet.

1. If a relation schema belongs to 2NF, then it _____.
A. may belong to 1NF B. certainly belongs to 1NF
C. certainly belongs to 3NF D. certainly belongs to BCNF
2. [☆] If all attributes of relation schema R are primary attributes, then R at least belongs to _____.
A. 1NF B. 2NF C. 3NF D. BCNF
3. If there is no partial FD in relation schema R that belongs to 1NF, then R certainly belongs to _____.
A. 1NF B. 2NF C. 3NF D. BCNF
4. A relation schema _____.
A. can have exactly one candidate key B. can have many primary keys
C. can have one or many candidate keys D. can have one or many primary keys
5. A candidate key can contain _____.
A. zero or many attributes B. no attribute
C. exactly one attribute D. one or many attributes
6. Given relation schema $R(S, D, M)$ and its FD set $F=\{S \rightarrow D, D \rightarrow M\}$, then at most R belongs to _____.
A. 1NF B. 2NF C. 3NF D. BCNF
7. Given relation schema $R(A, B, C, D)$ and its FD set $F=\{AB \rightarrow C, C \rightarrow D\}$, then at most R belongs to _____.
A. 1NF B. 2NF C. 3NF D. BCNF
8. If _____, then we call $X \rightarrow Y$ trivial FD.
A. $X \subseteq Y$ B. $Y \subseteq X$ C. $X \cap Y = \emptyset$ D. $X \cap Y \neq \emptyset$
9. In relation schema R , $X \rightarrow Y$ means that _____.
A. In one of R 's relation, if there are two rows that equal on X , then they also equal on Y .
B. In each of R 's relation, if there are two rows that equal on X , then they also equal on Y .
C. In one of R 's relation, values of Y equal values of X .
D. In each of R 's relation, values of Y equal values of X .
10. [☆] One disadvantage of decomposing relation schema is that _____.
A. data cannot be processed in a centralized way
B. we may lose some data
C. we may encounter operation anomalies
D. the time-consuming join operation is often needed in most queries

PART B

Fill in blanks with proper words or phrases.

1. If $X \rightarrow Y$ and every real subset of X does not functionally determine Y , then we call $X \rightarrow Y$ a (a); otherwise, If $X \rightarrow Y$ and one real subset of X also functionally determine Y , then we call $X \rightarrow Y$ a (b).
2. In relational model, every relation schema must at least belong to (a).

3. If relation schema $R \in 1NF$ and each non-primary attribute of R is (a) a candidate key of R , then $R \in 2NF$.
4. If relation schema $R \in 1NF$ and each non-primary attribute of R is (a) any candidate key of R , then $R \in 3NF$.
5. Given relation schema $R(A,B,C,D,E, F, G)$ and its FD set $F=\{A \rightarrow B, A \rightarrow C, C \rightarrow D, C \rightarrow E, A \rightarrow F, F \rightarrow G\}$, then the candidate key(s) of R will be (a); at most R 's normal form is (b).
6. Given relation schema $R(A, C, D)$ and its FD set $F=\{A \rightarrow C, A \rightarrow D\}$, then the candidate key(s) of R will be (a); at most R 's normal form is (b).